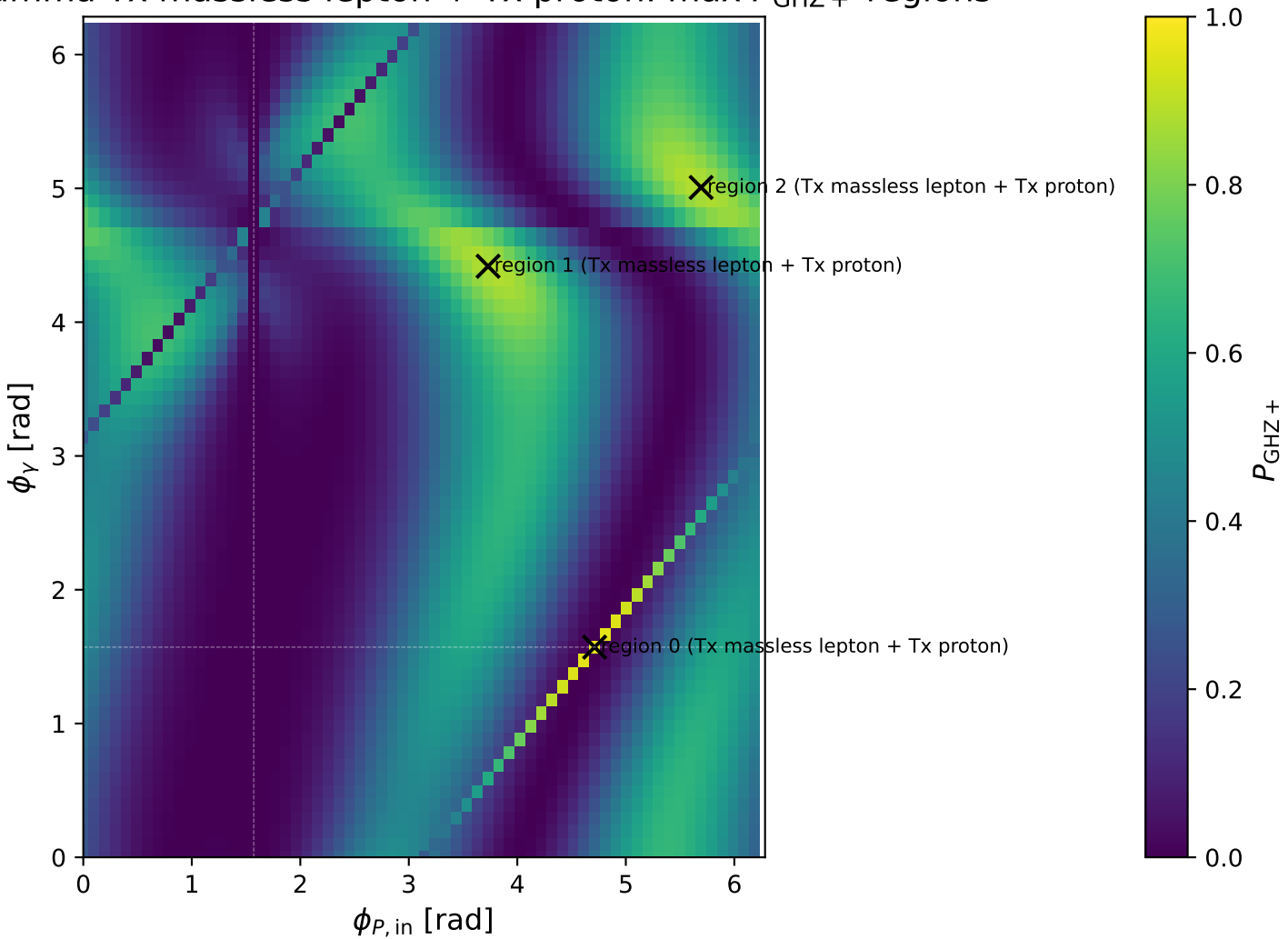
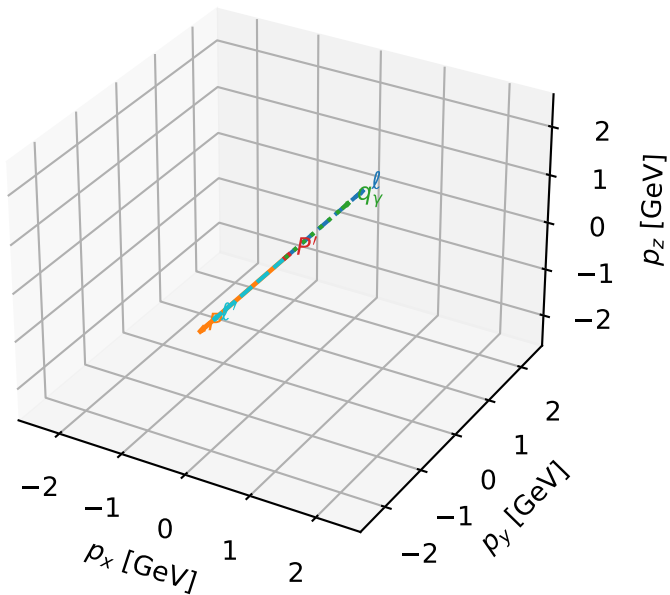


gamma Tx massless lepton + Tx proton: max $P_{\text{GHz}+}$ regions



Tx massless lepton + Tx proton characteristic max P_{GHZ} + configuration

3D momenta



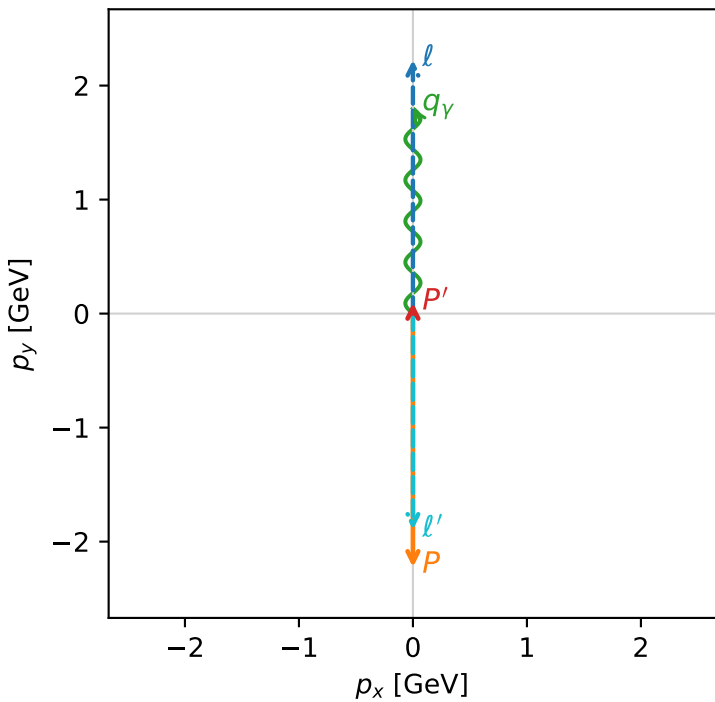
ghz_purity_high_egamma_txtx_region_0 $P_{\text{GHZ}}=0.964874$
 region: high_Egamma_TxTx_region_0
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_high_Egamma
 incoming state: $|T_{x,\ell_0}\rangle \otimes |T_{x,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=0.0956529$
 $\theta_{\text{in}}=1.57079$, $\phi_l=1.5708$, $\phi_p=4.71239$
 $E_\gamma=1.8$, $\phi_\gamma=1.5708$
 $Q^2=16.8669$, $x_B=0.846865$, $t=-3.21819$, $W^2=3.92981$, $y=0.965151$
 $\theta(\ell', \gamma)=180$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 2.2244$ $p_x= 0.0000$ $p_y= 2.2244$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= -0.0000$ $p_y= -2.2244$ $p_z= 0.0000$
 ℓ' $E= 1.8957$ $p_x= -0.0000$ $p_y= -1.8957$ $p_z= 0.0000$
 P' $E= 0.9429$ $p_x= 0.0000$ $p_y= 0.0957$ $p_z= 0.0000$
 q_γ $E= 1.8000$ $p_x= 0.0000$ $p_y= 1.8000$ $p_z= 0.0000$

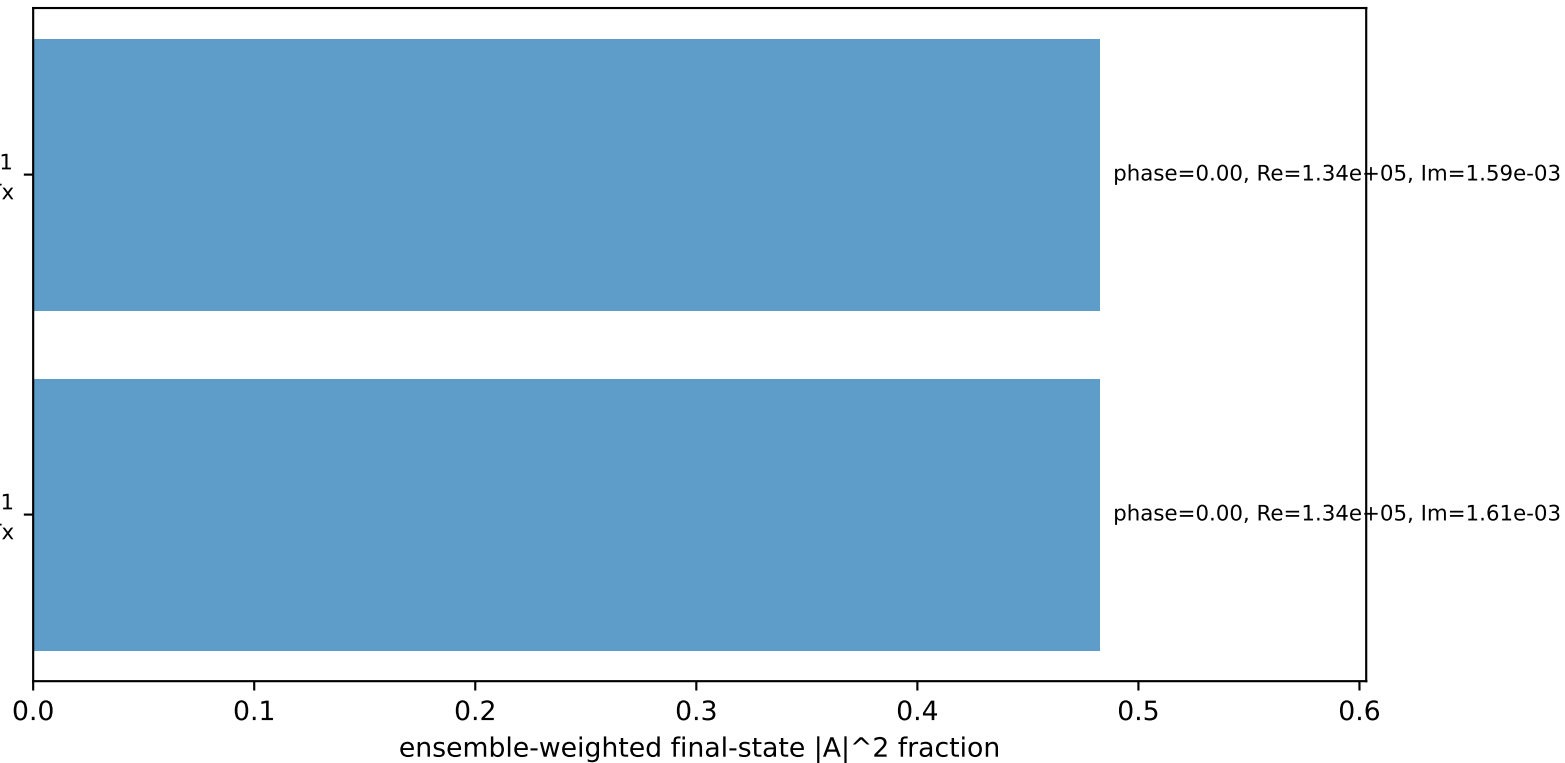
Transverse plane



$h_e=-1, h_p=-1, h_\gamma=-1$
 electron Tx, proton Tx

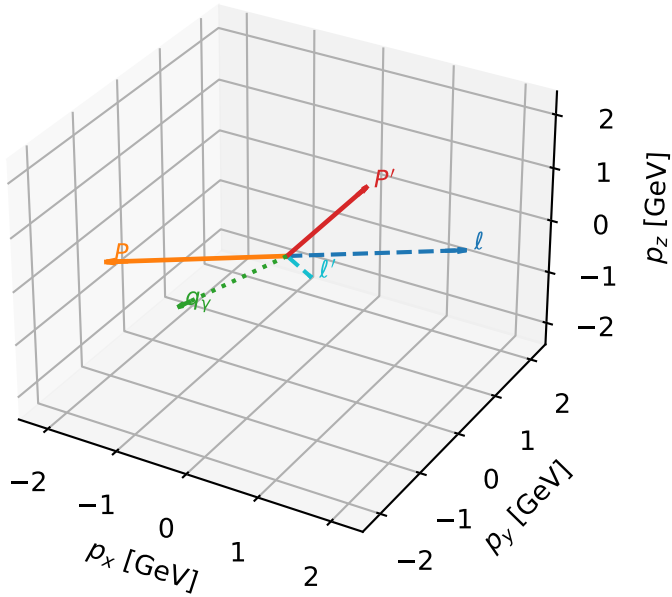
$h_e=+1, h_p=+1, h_\gamma=+1$
 electron Tx, proton Tx

Leading initial-ensemble/final-state components (N=2, retained=96.5%)



Tx massless lepton + Tx proton characteristic max P_{GHZ} + configuration

3D momenta

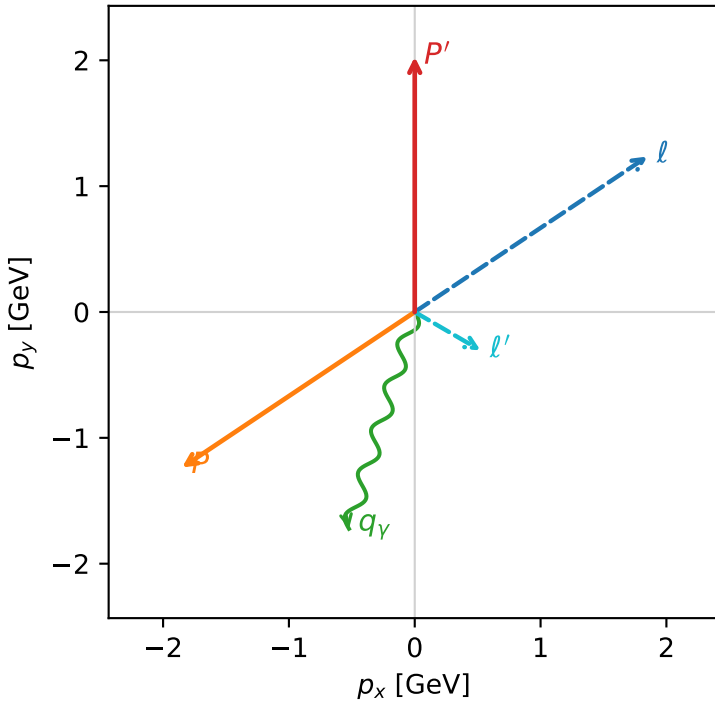


ghz_purity_high_egamma_txtx_region_1 $P_{\text{GHZ}}=0.885903$
 region: high_Egamma TxTx_region_1
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_high_Egamma
 incoming state: $|T_{x,\ell_0}\rangle \otimes |T_{x,p}\rangle$

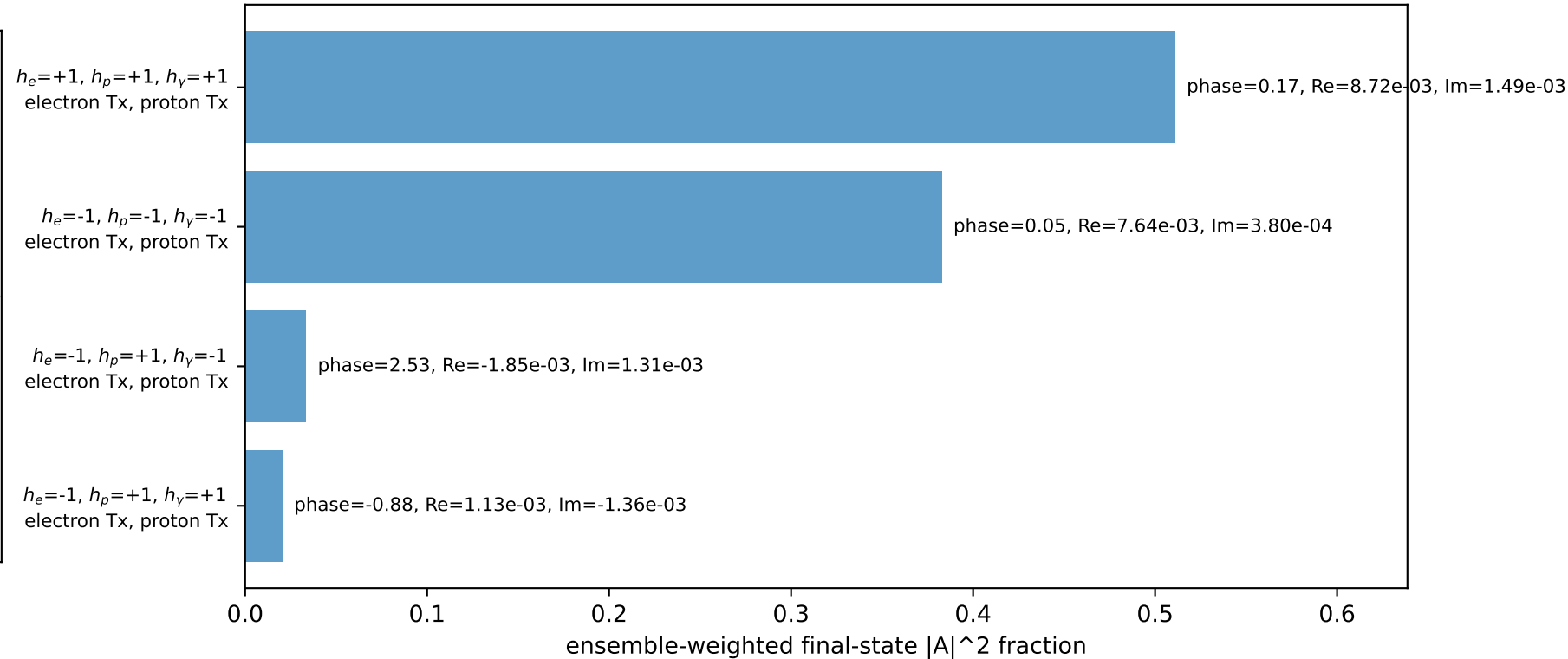
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=2.02717$
 $\theta_{\text{in}}=1.57079$, $\phi_{\ell}=0.589049$, $\phi_p=3.73064$
 $E_{\gamma}=1.8$, $\phi_{\gamma}=4.41786$
 $Q^2=1.51113$, $x_B=0.0913849$, $t=-14.0353$, $W^2=15.9046$, $y=0.801313$
 $\theta(\ell', \gamma)=76.6286$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= 1.8495$ $p_y= 1.2358$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= -1.8495$ $p_y= -1.2358$ $p_z= 0.0000$
 ℓ' $E= 0.6049$ $p_x= 0.5225$ $p_y= -0.3047$ $p_z= 0.0000$
 P' $E= 2.2337$ $p_x= 0.0000$ $p_y= 2.0272$ $p_z= 0.0000$
 q_{γ} $E= 1.8000$ $p_x= -0.5225$ $p_y= -1.7225$ $p_z= 0.0000$

Transverse plane

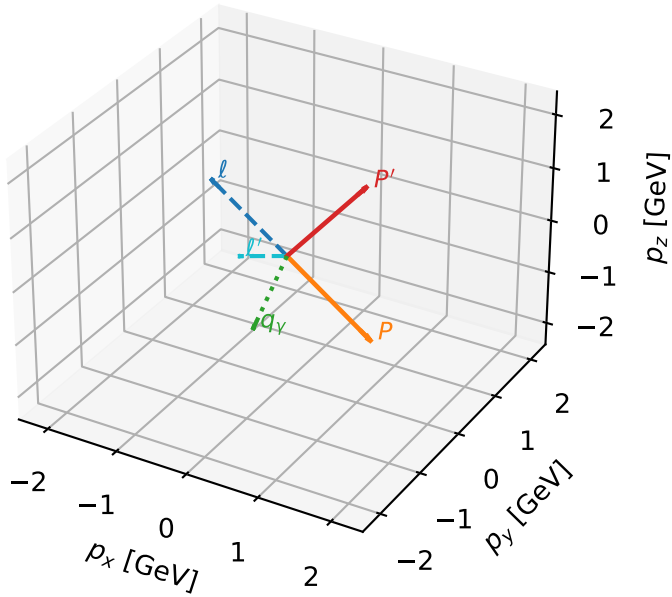


Leading initial-ensemble/final-state components (N=4, retained=94.8%)



Tx massless lepton + Tx proton characteristic max P_{GHZ} + configuration

3D momenta



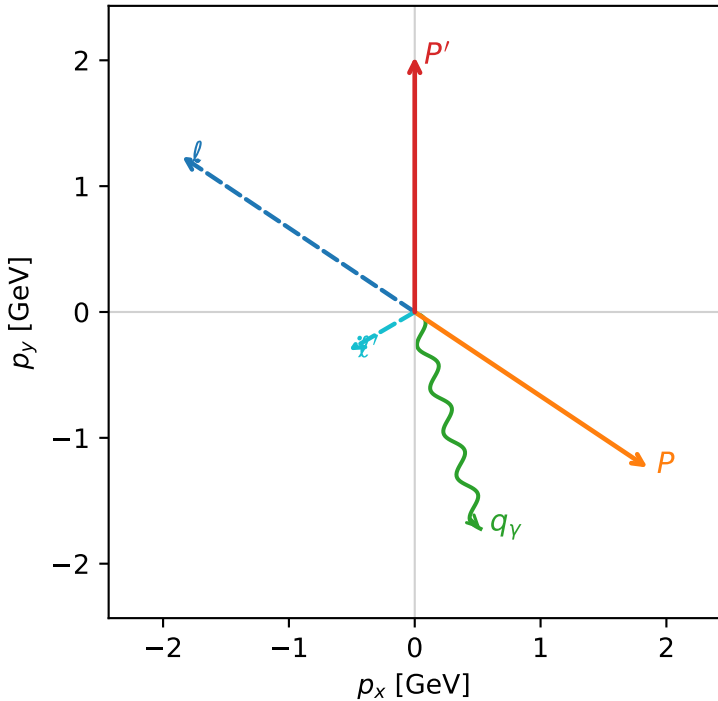
ghz_purity_high_egamma_txtx_region_2 $P_{\text{GHZ}}=0.885903$
 region: high_Egamma TxTx_region_2
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_high_Egamma
 incoming state: $|T_{x,\ell_0}\rangle \otimes |T_{x,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=2.02717$
 $\theta_{\text{in}}=1.57079$, $\phi_\ell=2.55254$, $\phi_p=5.69414$
 $E_\gamma=1.8$, $\phi_\gamma=5.00691$
 $Q^2=1.51113$, $x_B=0.0913849$, $t=-14.0353$, $W^2=15.9046$, $y=0.801313$
 $\theta(\ell', \gamma)=76.6286$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 2.2244$ $p_x= -1.8495$ $p_y= 1.2358$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 1.8495$ $p_y= -1.2358$ $p_z= 0.0000$
 ℓ' $E= 0.6049$ $p_x= -0.5225$ $p_y= -0.3047$ $p_z= 0.0000$
 P' $E= 2.2337$ $p_x= 0.0000$ $p_y= 2.0272$ $p_z= 0.0000$
 q_γ $E= 1.8000$ $p_x= 0.5225$ $p_y= -1.7225$ $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=94.8%)

